

BAAZ JKBOSE

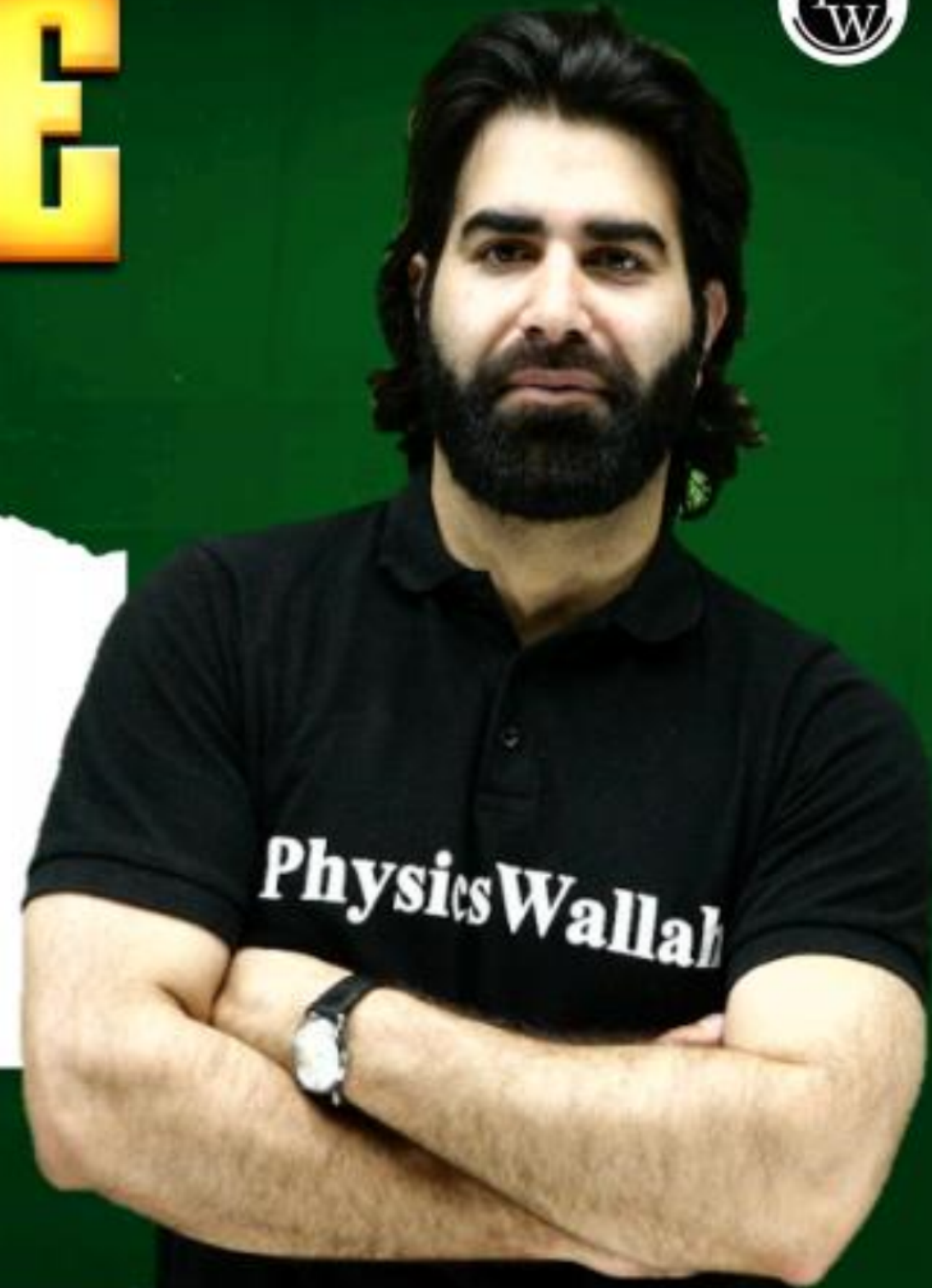
2026 CLASS-11TH

Body Fluids and Circulation

LECTURE-06

Zoology

Dr Adil Altaf sir



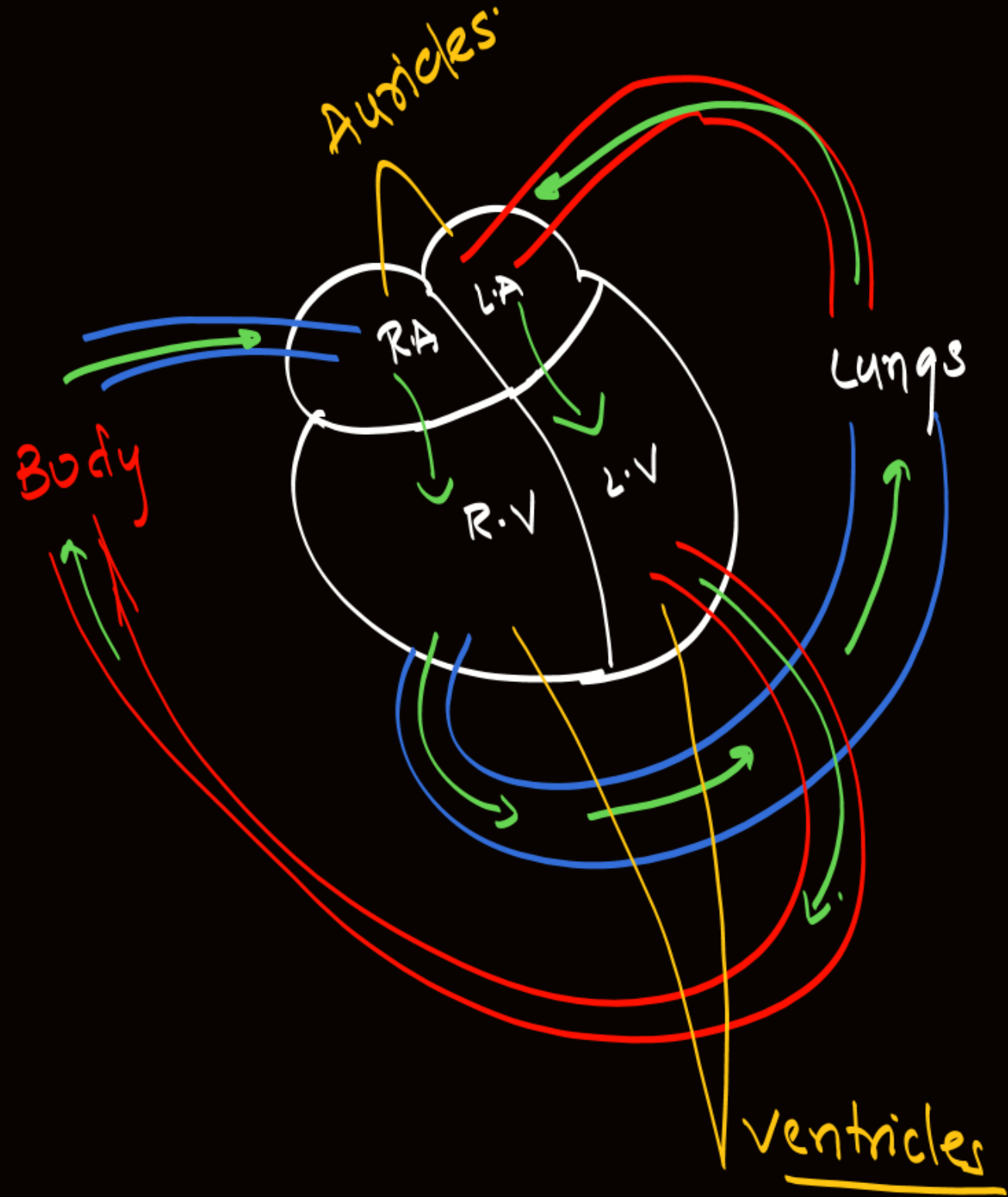
Topics To Be Covered

1 Cardiac Cycle, ECG.

Revision

→ Human Heart

4 chamber



Cardiac Cycle

↳ The are sequence of Events which are repeated cyclically. (in cyclic Manner) is k/as Cardiac cycle.

[Joint Diastole]

Both ventricles & Auricles are at Relaxation.

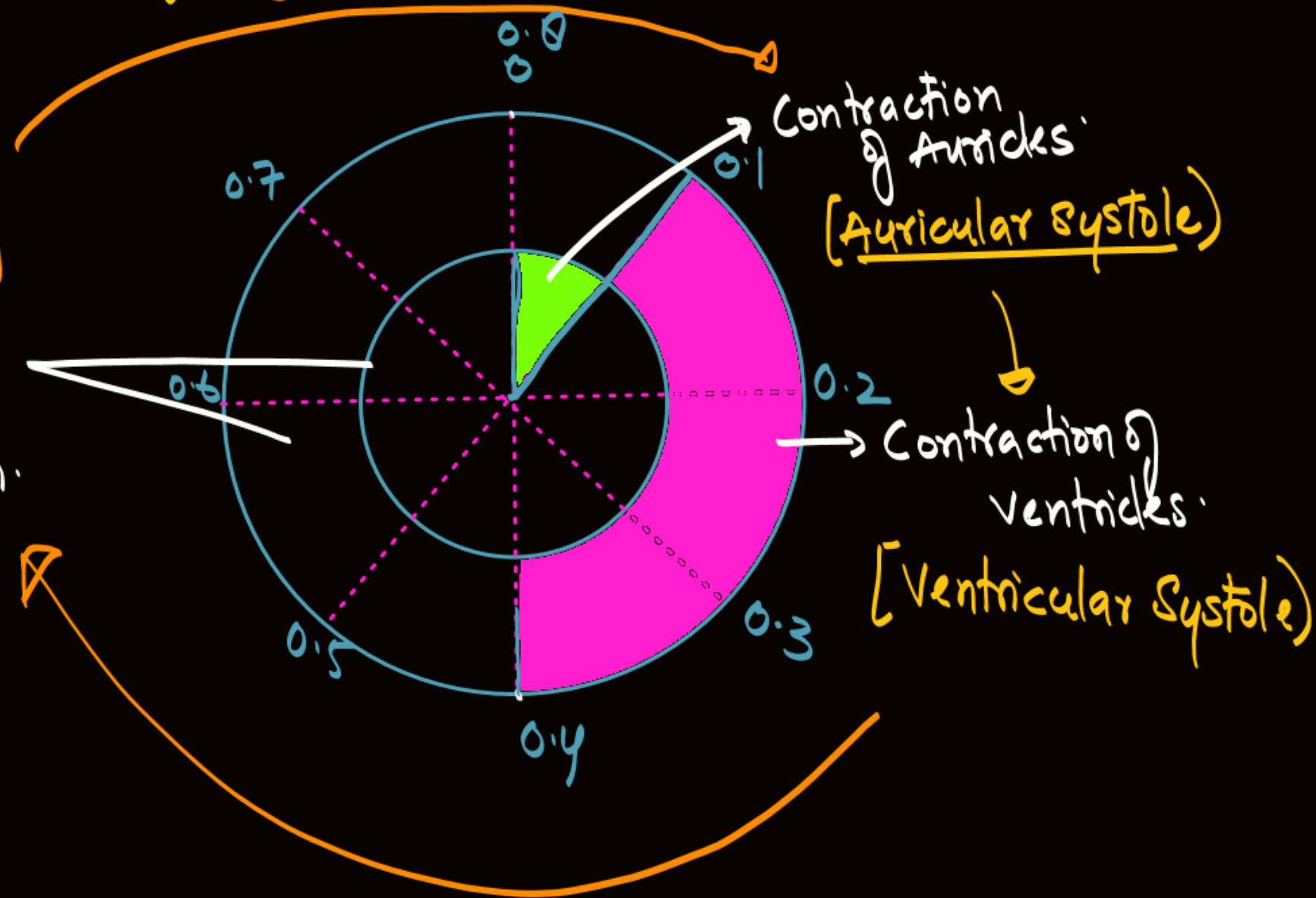


Diagram:

- Inner circle = Auricles
- Outer circle = Ventricles

- Contraction = Systole
- Relaxation = Diastole

Events in cardiac cycle

- (a) Auricular Systole
- (b) Ventricular Systole
- (c) Joint Diastole

Time period of Cardiac Cycle

$$\text{Time period} = \frac{1}{\text{Frequency}} \quad \text{--- ①}$$

$$\text{Frequency} = \frac{\text{No. of cycles}}{\text{Time in seconds}}$$

$$\text{Time Period} = \frac{\text{Time in seconds}}{\text{No. of cycles}}$$

e.g. ^{Normal} Average Heart Beat (70-75 times/Min) ^(72 bpm)

$$\text{Time period} = \frac{60 \text{ seconds (Given time)}}{75 \text{ cycles (cycle)}}$$

$$\text{Time period} = \underline{0.8 \text{ second}} \rightarrow \underline{\text{Average}}$$

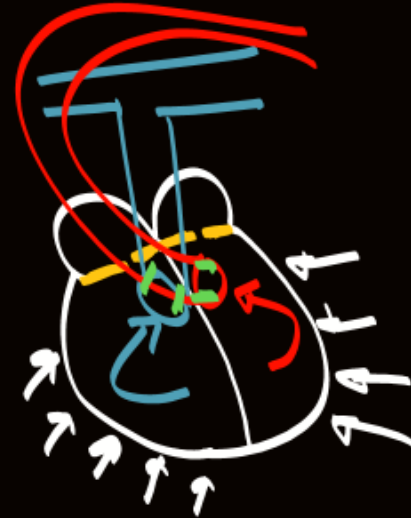
Q No 1 → If a person's [while playing football] Heart Rate is 150 Beats/Min. Find Time period of Cardiac cycle?

$$\text{Time period} = \frac{\text{Given time}}{\text{cycles}} = \frac{60 \text{ second}}{150} = \underline{0.4 \text{ seconds}}$$

In Detail.

Joint Diastole (0.4sec)

- All 4 chambers at Rest
- A.V valves are open.
- Auricles Receiving Blood.
- 70% of ventricles get Filled.



Auricular (Atrial) Systole (0.1sec)

- SA Node generates Signal.
- Causes contraction of Auricles
- ventricles Fully Filled with Blood because Auricles pump Blood in ventricles.

Ventricular Systole [0.3 sec]. (-Beginning-)

- AV Node generates Impulse/Signal.
- Contraction of ventricles.
- Closure of AV valves [First Heart Sound = Lubb].
- Opening of Semilunar Valves.
- Blood pumped in P. Artery & Aorta.

End

- Relaxation of ventricles.
- Closure of Semilunar Valves [Second Heart Sound = Dubb].
- Opening of AV valves.

⇒ Parameters in Cardiac cycle

① Heart Rate [Normal = 70 - 75 bpm].
Average = 72 bpm.

② Stroke volume

Amount of Blood Pumped by Heart (left ventricle) in one beat

$$S.V = 70 - 80 \text{ ml.}$$

③ Cardiac Output

Amount of Blood Pumped by Heart (left ventricle) in one Minute.

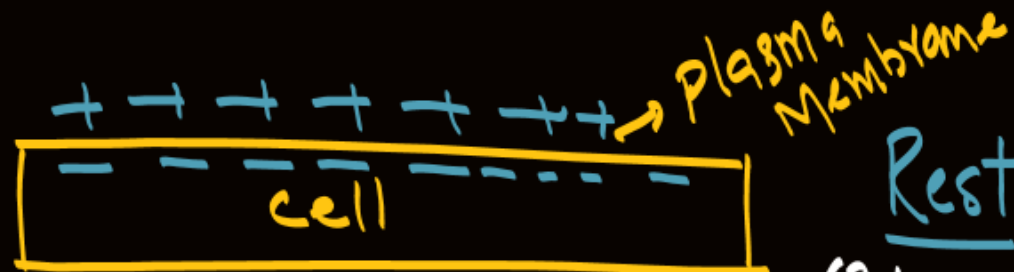
$$\text{Cardiac output} = S.V \times \text{Heart Rate}$$

$$\downarrow \text{Average} = 70 \times 75 \quad \approx 5000 \text{ ml.}$$

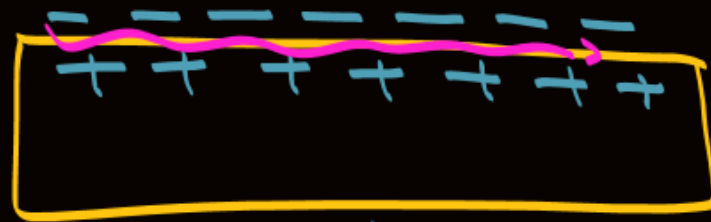
$$\approx \underline{5 \text{ litre}}$$

Extra

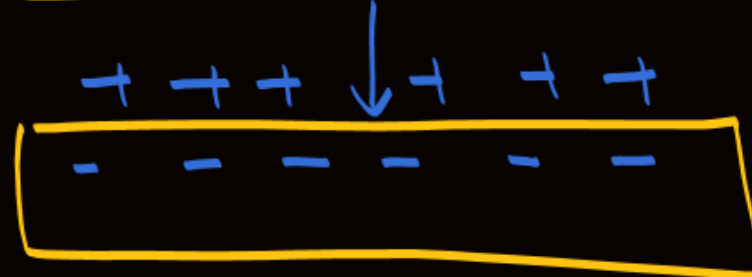
Muscle (cell - polarised) $\rightarrow$ Charge difference



Rest - state - polarisation
(Relaxation, Diastole)



Contraction - Depolarization
(Systole)



Rest = (Repolarization)

ECG.

↳ Electrocardiogram (Report)

- It is a graphical Representation of Electrical Activity of Heart

↳ Machine = Electrocardiograph.

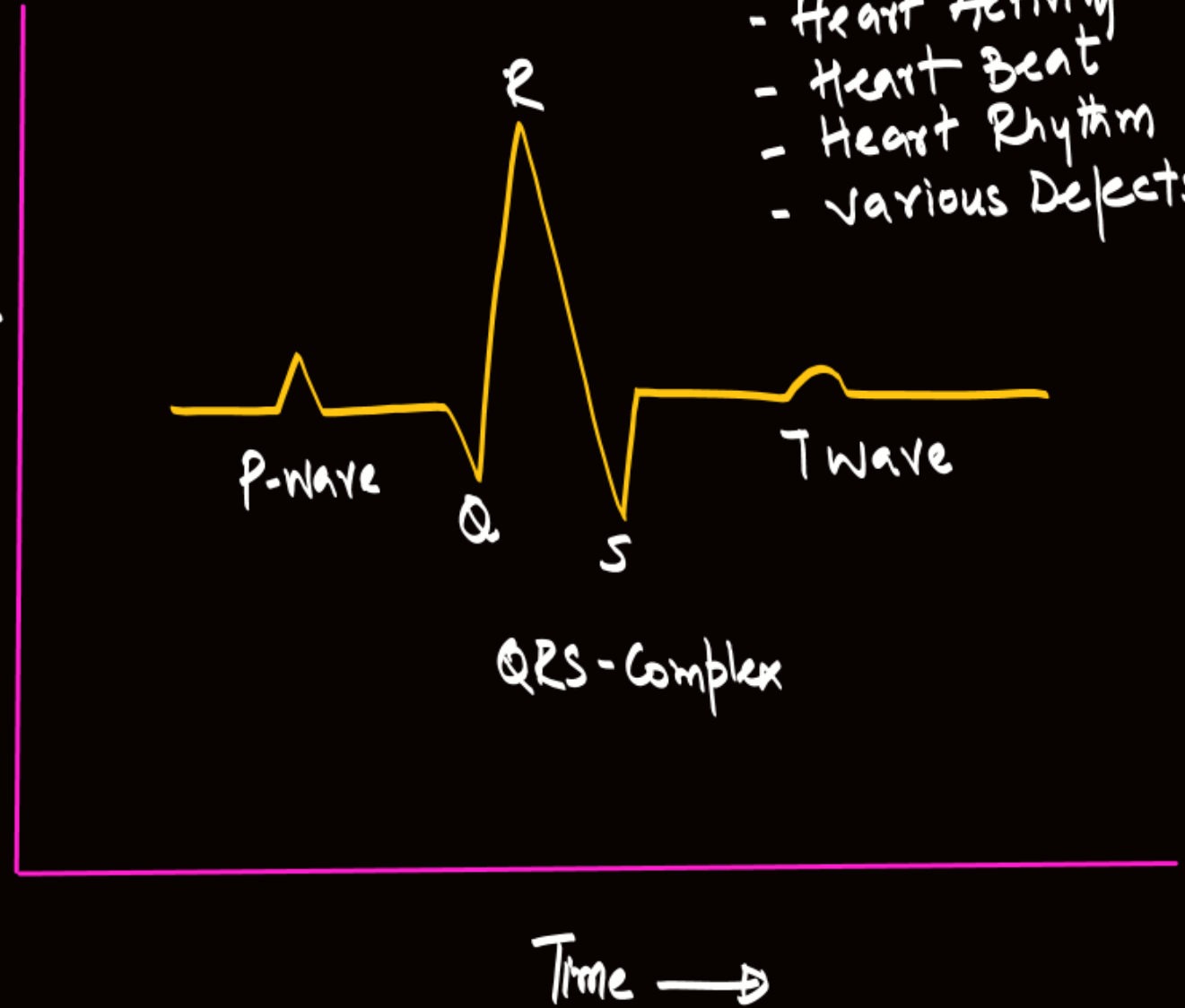
↳ 3 Main Leads, one on each wrist, and one on left Ankle:

↳ 6 chest Leads.

↳ In Standard ECG u will Find,

- ⇒ P wave = Auricular depolarisation [Auricular Systole].
- ⇒ QRS complex = Ventricular depolarisation [ventricular Systole].
- ⇒ T wave = Ventricular Repolarisation [ventricular Diastole].

(mV) Voltage
↑

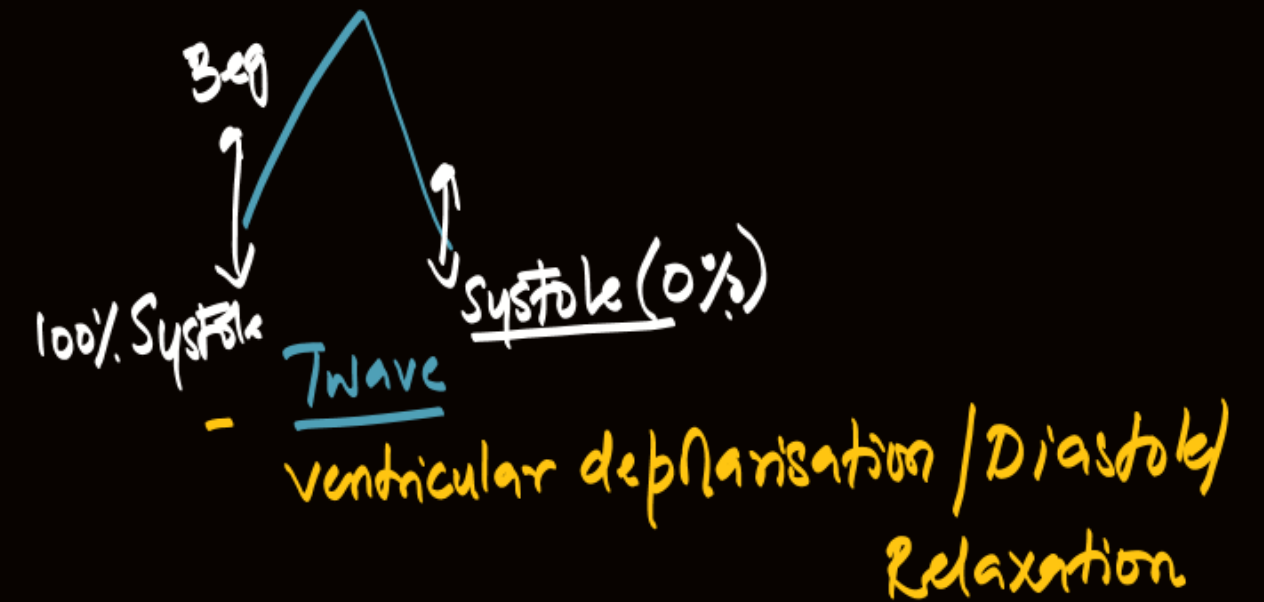


USES of ECG

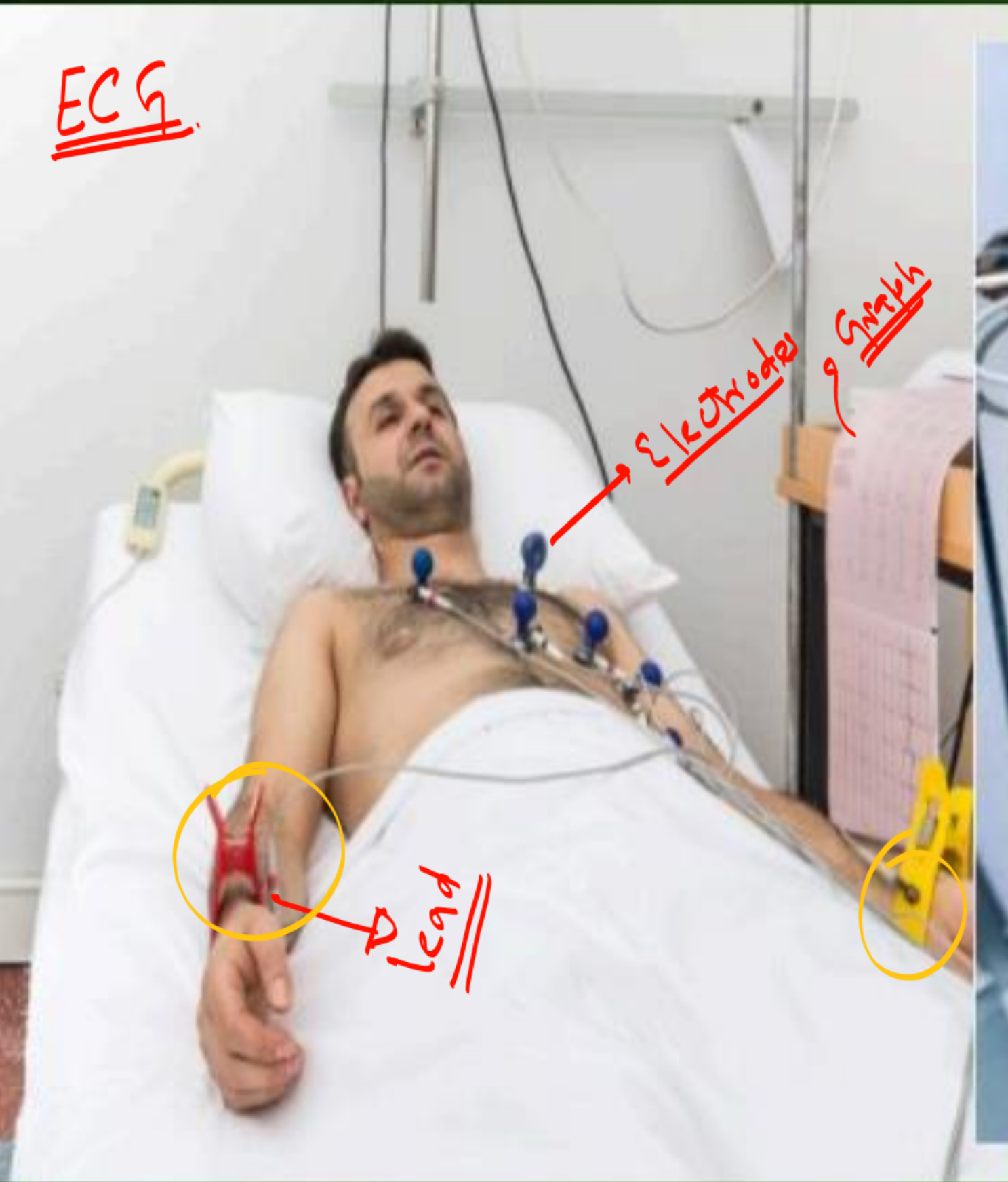
- Heart Activity
- Heart Beat
- Heart Rhythm
- Various Defects.

Standard ECG.

End of T-wave Marks end of ventricular Systole



ECG.



Thank
You